Theorem 1.12

The rank of a non-zero matrix is equal to the number of non-zero rows in a row-echelon form of the matrix.

Example 1.17

Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$ by reducing it to a row-echelon form.

Solution

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$. Applying elementary row operations, we get

$$A \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -2 \\ 0 & -6 & -4 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The last equivalent matrix is in row-echelon form. It has two non-zero rows. So, $\rho(A) = 2$.

Example 1.18

Find the rank of the matrix $\begin{bmatrix} 2 & -2 & 4 & 3 \\ -3 & 4 & -2 & -1 \\ 6 & 2 & -1 & 7 \end{bmatrix}$ by reducing it to an echelon form.

Solution

Let A be the matrix. Performing elementary row operations, we get

$$A = \begin{bmatrix} 2 & -2 & 4 & 3 \\ -3 & 4 & -2 & -1 \\ 6 & 2 & -1 & 7 \end{bmatrix} \xrightarrow{R_2 \to 2R_2} \begin{bmatrix} 2 & -2 & 4 & 3 \\ -6 & 8 & -4 & -2 \\ 6 & 2 & -1 & 7 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 + 3R_1} \begin{bmatrix} 2 & -2 & 4 & 3 \\ 0 & 2 & 8 & 7 \\ 0 & 8 & -13 & -2 \end{bmatrix}.$$

$$\xrightarrow{R_3 \to R_3 - 4R_2} \begin{bmatrix} 2 & -2 & 4 & 3 \\ 0 & 2 & 8 & 7 \\ 0 & 0 & -45 & -30 \end{bmatrix} \xrightarrow{R_3 \to R_3 \div (-15)} \begin{bmatrix} 2 & -2 & 4 & 3 \\ 0 & 2 & 8 & 7 \\ 0 & 0 & 3 & 2 \end{bmatrix}.$$

The last equivalent matrix is in row-echelon form. It has three non-zero rows. So, $\rho(A) = 3$.

Elementary row operations on a matrix can be performed by pre-multiplying the given matrix by a special class of matrices called elementary matrices.

Definition 1.7

An **elementary matrix** is defined as a matrix which is obtained from an identity matrix by applying only one elementary transformation.

Remark

If we are dealing with matrices with three rows, then all elementary matrices are square matrices of order 3 which are obtained by carrying out only one elementary row operations on the unit matrix I_3. Every elementary row operation that is carried out on a given matrix A can be obtained by pre-multiplying A with elementary matrix. Similarly, every elementary column operation that is carried out on a given matrix A can be obtained by post-multiplying $A$with an elementary matrix. In the present chapter, we use elementary row operations only.

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